Course Title: Ethical Hacking

Objective :

- To install and configure Kali Linux, Metasploitable 2, and Windows XP in VirtualBox.
- To create a virtualized cyber security lab environment.
- To establish network connectivity between virtual machines for security testing.

Methodology :

1. Install VirtualBox on the host system.
2. Create virtual machines for Kali Linux and Windows XP.
3. Import the Metasploitable 2 virtual machine.
4. Configure memory, storage, and network settings.
5. Install Kali Linux and Windows XP using their respective ISO files.
6. Start all virtual machines and verify successful installation.
7. Configure networking and test connectivity between the systems.

Expected Outcome :

- Kali Linux, Metasploitable 2, and Windows XP are successfully installed and operational in VirtualBox.
- A functional cyber security laboratory environment is established.

Guide to install Kali Linux, metasploitable2 and Windows XP

Introduction to kali linux :

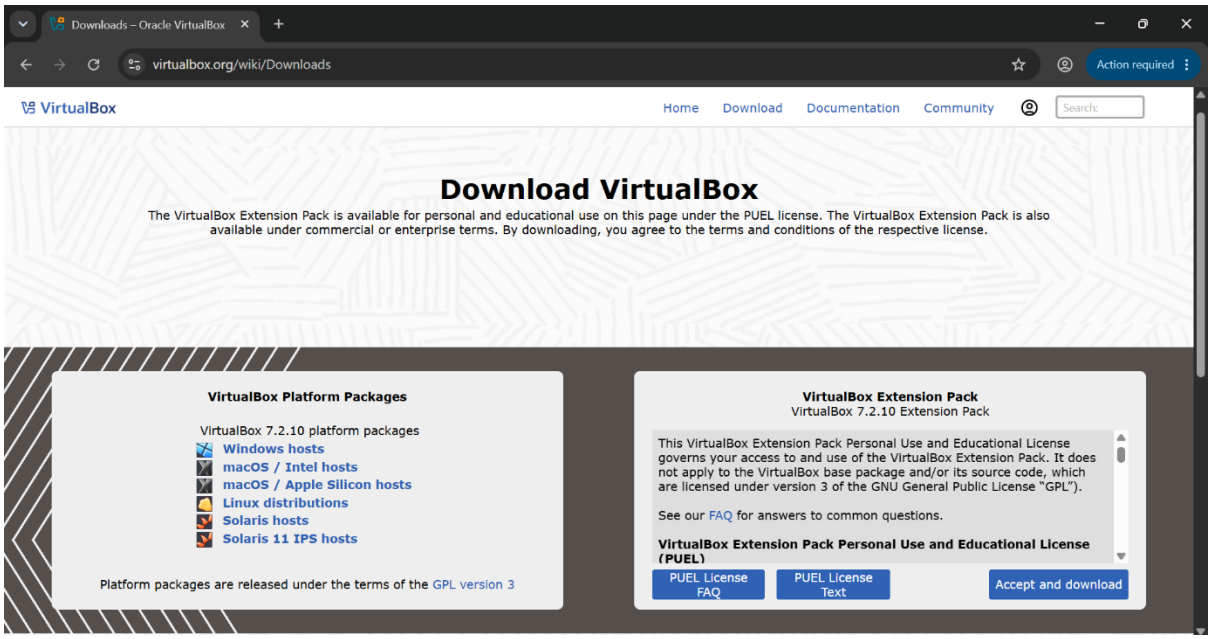
Kali Linux is a Debian-based Linux distribution developed and maintained by the Offensive Security team. It is specifically designed for Cyber Security professionals, Ethical Hackers, Penetration Testers, Digital Forensics Investigators, and Security Researchers. Kali Linux comes pre-installed with hundreds of security tools.

Installation of Virtual Box :

Step 1: Open the VirtualBox Website

Go to:

Oracle VM VirtualBox Downloads

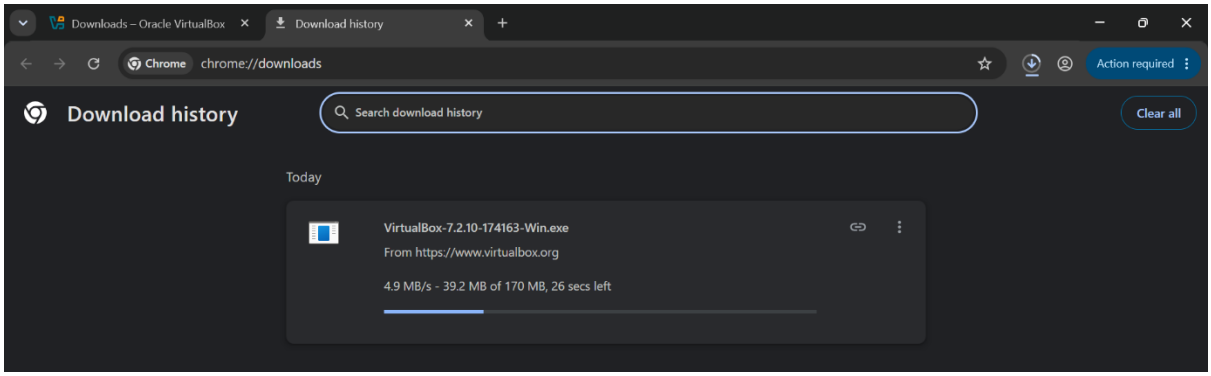


Step 2: Download VirtualBox

Under VirtualBox Platform Packages, click:

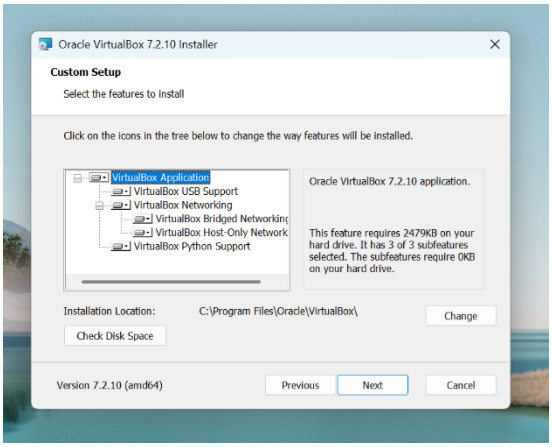
- Windows hosts (if your computer runs Windows)

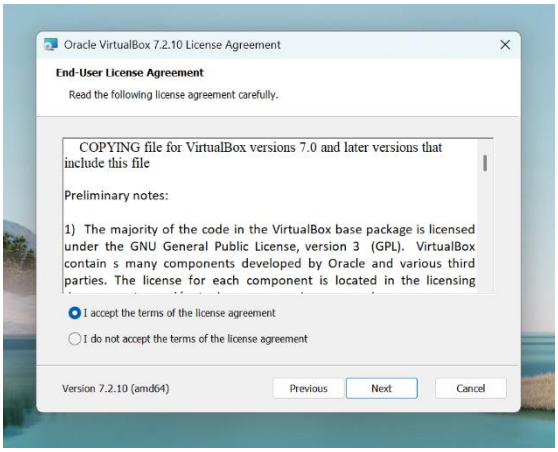
This will download the VirtualBox installer (VirtualBox-x.x.x-Win.exe).

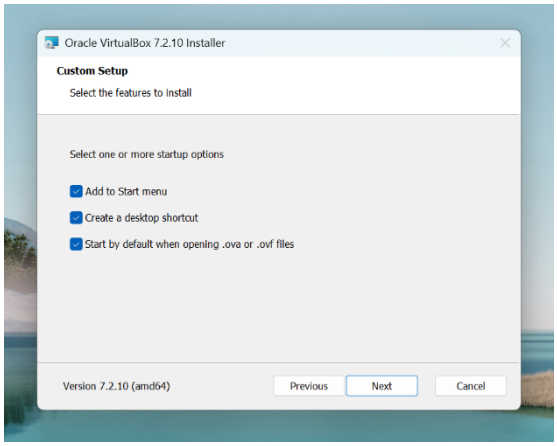


Step 3: Run the Installer

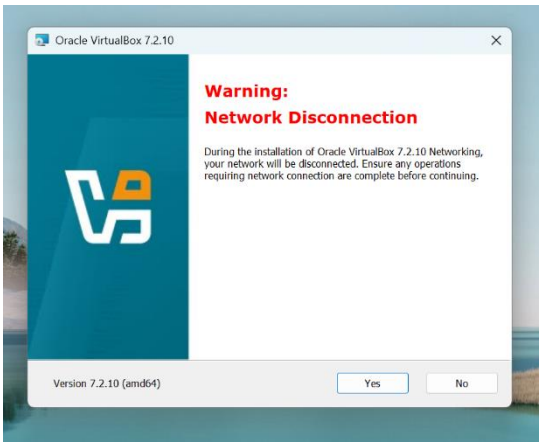
1. Double-click the downloaded file.
2. Click Next.
3. Keep the default settings.
4. Click Next.







5. Click Yes if a network warning appears.



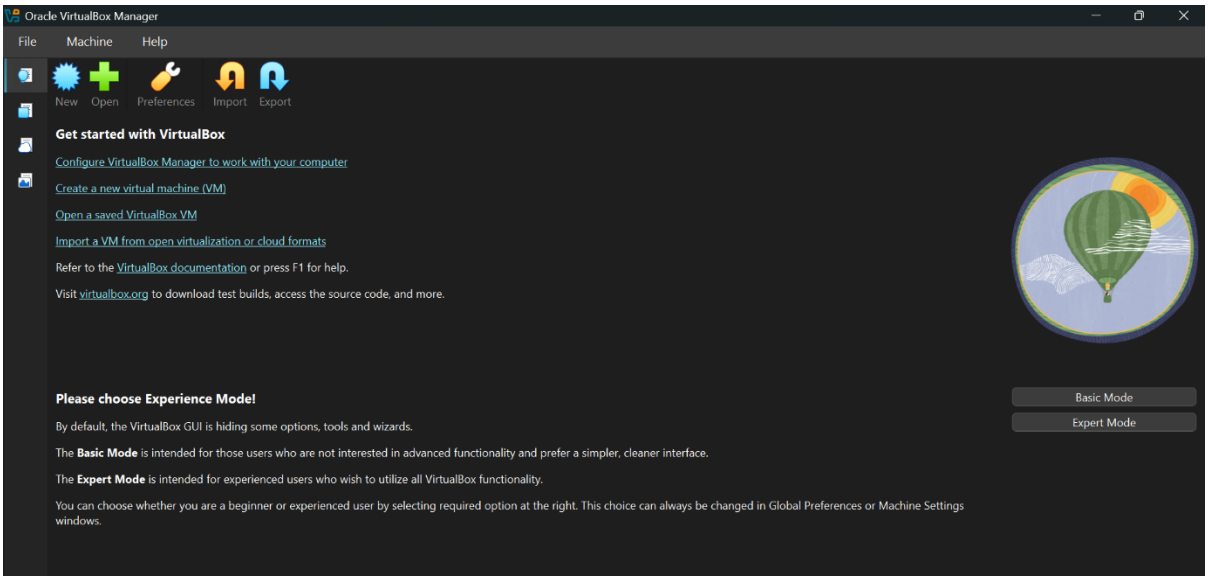
6. Click Install.
7. Allow administrator permission if prompted.
8. Click Finish after installation completes.



Step 4: Verify Installation

1. Open VirtualBox.
2. You should see the VirtualBox Manager window.

3. Click New to create a virtual machine.

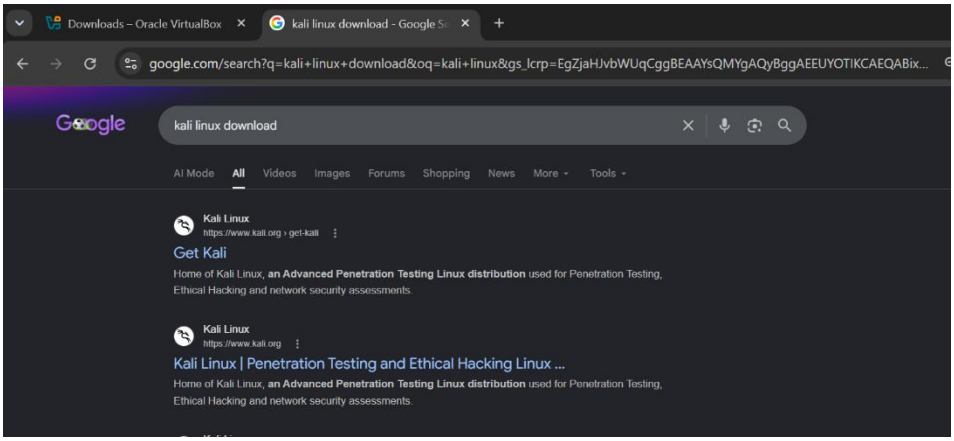


Installation and Setting up Kali Linux :

Step 1: Visit Official Website

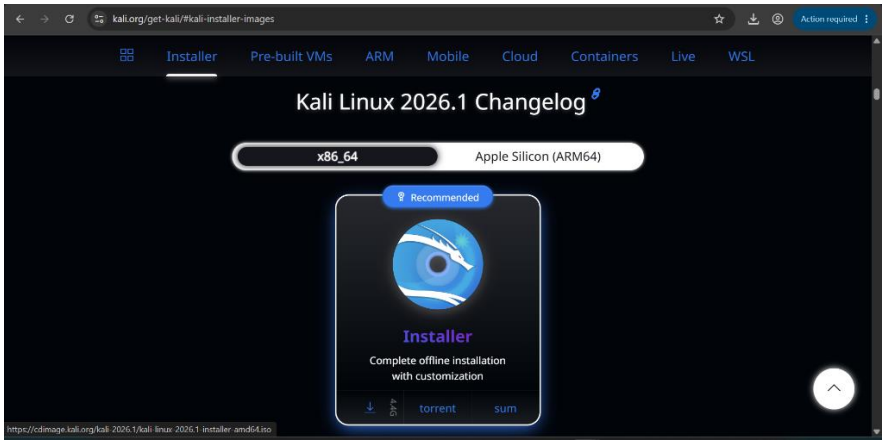
Open a web browser and search for download Kali Linux.

https://www.kali.org



Step 2: Download Installer Image

1. Select Installer Images
2. Select Installer x86 (64-bit)
3. Download the latest stable version

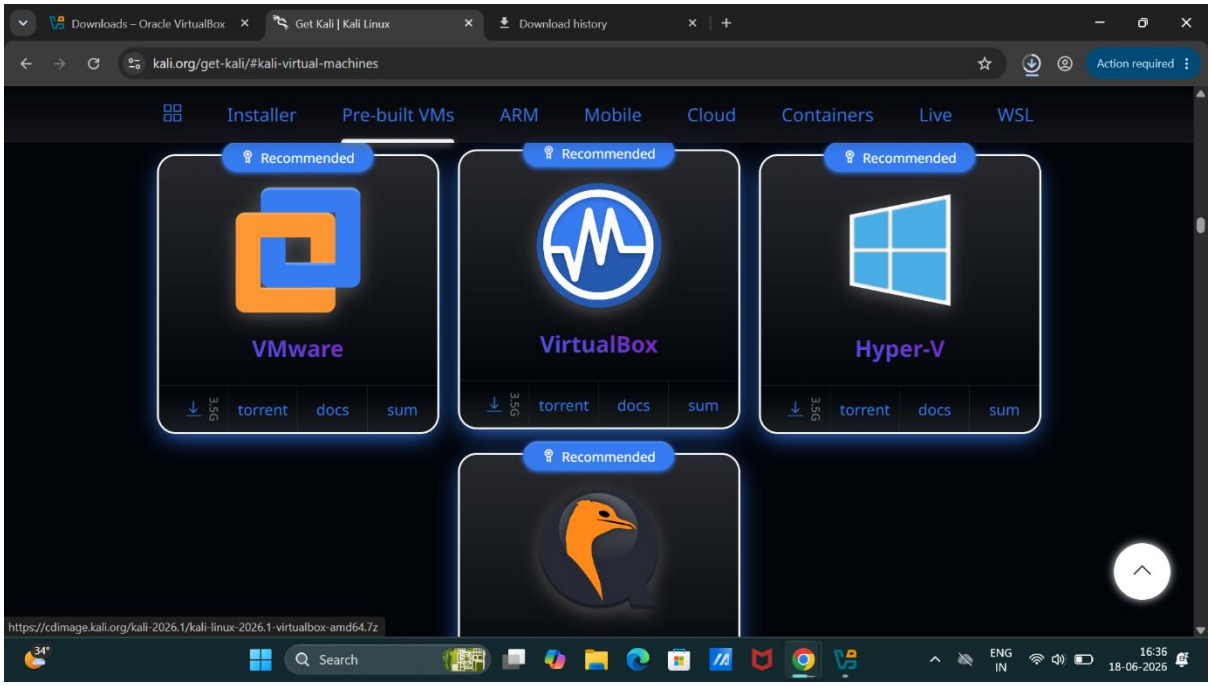


Installing Kali Linux in VirtualBox :

Step 1: Install VirtualBox

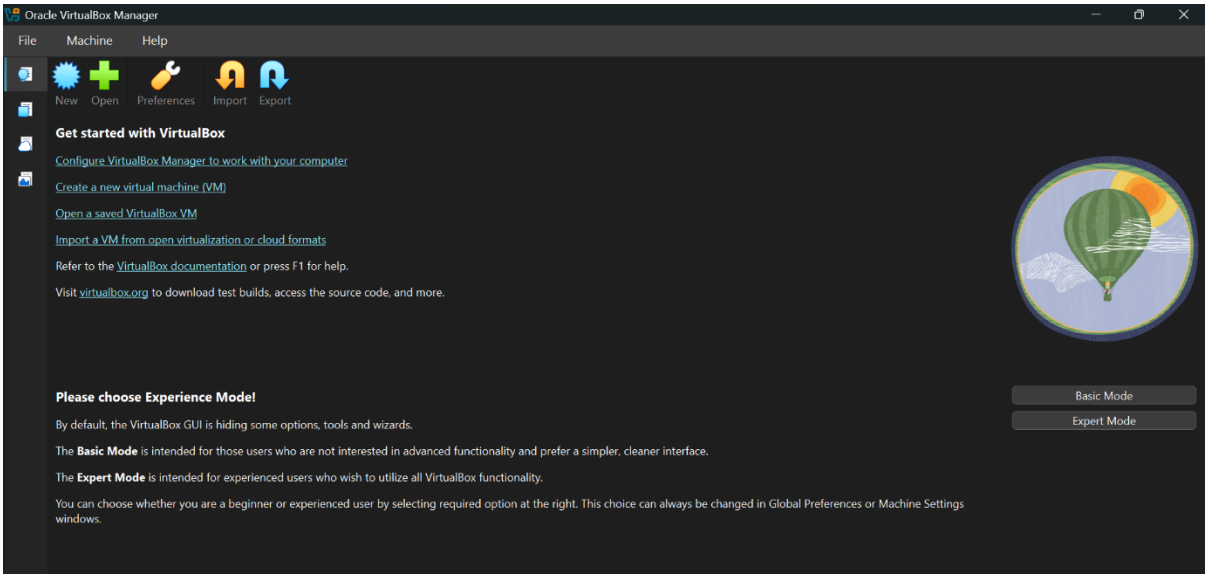
Download and install Oracle VirtualBox from:

https://www.virtualbox.org

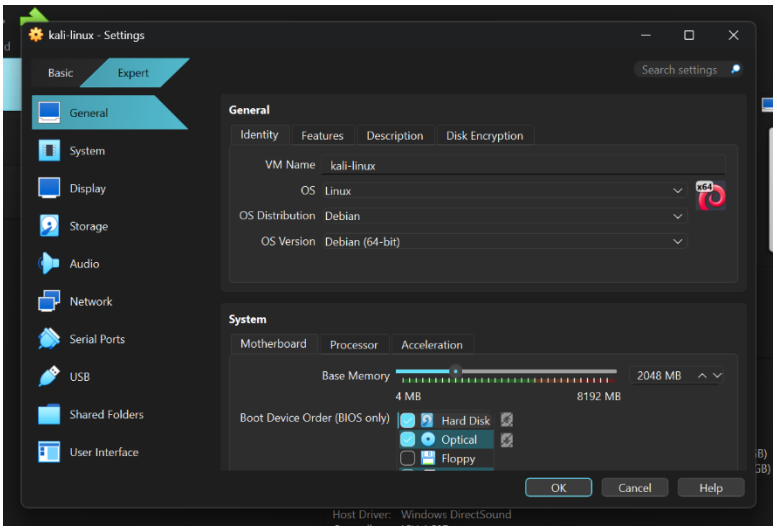


Step 2: Create a New Virtual Machine

1. Open VirtualBox.
2. Click New.



3. Enter VM Name:
    - Kali Linux
4. Type:
    - Linux
5. Version:
    - Debian (64-bit)
6. Click Next.



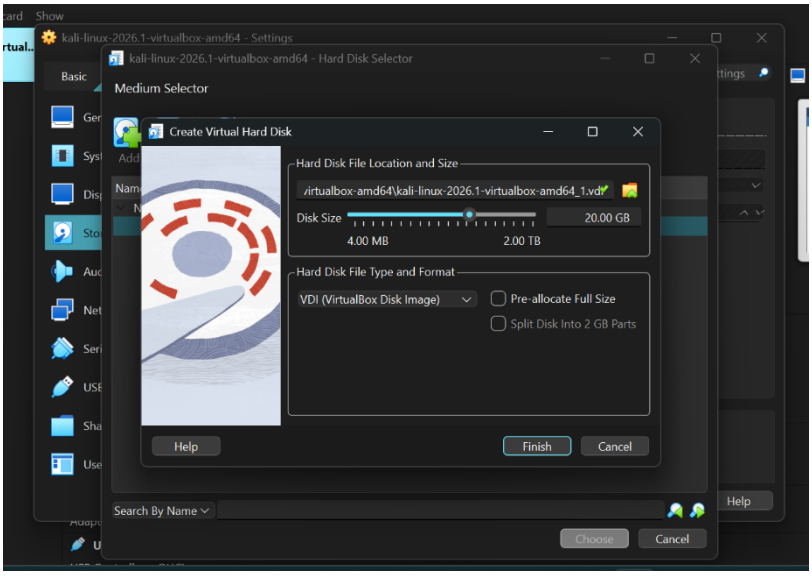
Step 3: Create Virtual Hard Disk

Click storage. Near Controller SATA, You can see add hard disk

1. Select:

- Create a Virtual Hard Disk Now

2. Click Create
3. Select:
    - VDI (VirtualBox Disk Image)
4. Click Next
5. Disk Size:
    - 20 GB or More
6. Click Create



Kali Linux Installation Procedure :

Step 1: Start Virtual Machine

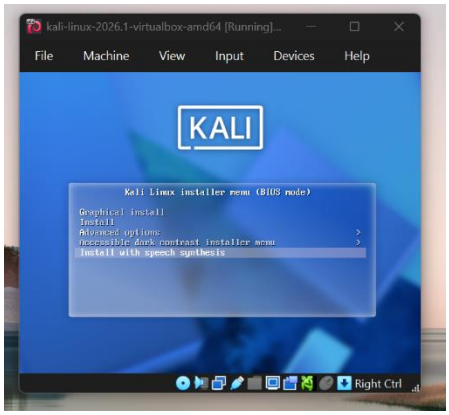
Click Start.

The Kali Linux boot menu appears.

Select:

Graphical Install

Press Enter.

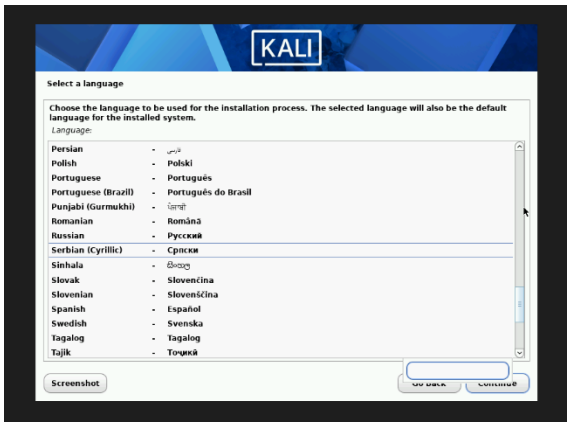


Step 2: Select Language

Choose:

English

Click Continue.



Step 3: Select Location

Choose your country.

Example:

India

Click Continue.



Step 4: Configure Keyboard

Select:

American English

Click Continue.



Step 5: Configure Network

Hostname:

kali



Domain Name:

Leave blank (optional)

Click Continue.



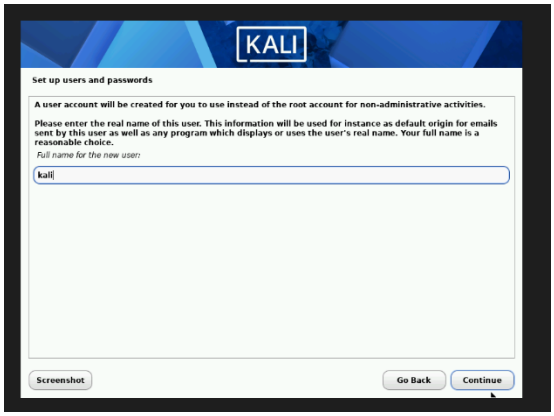
Step 6: Create User Account

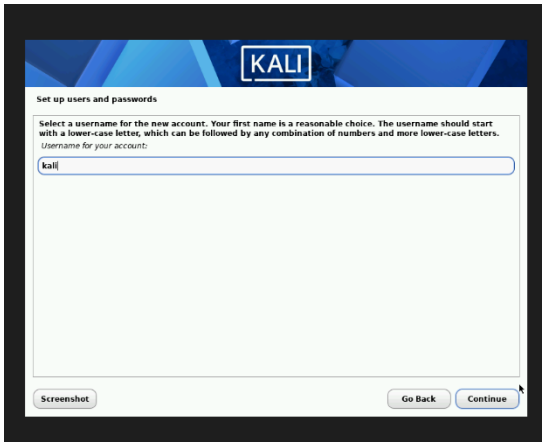
Enter:

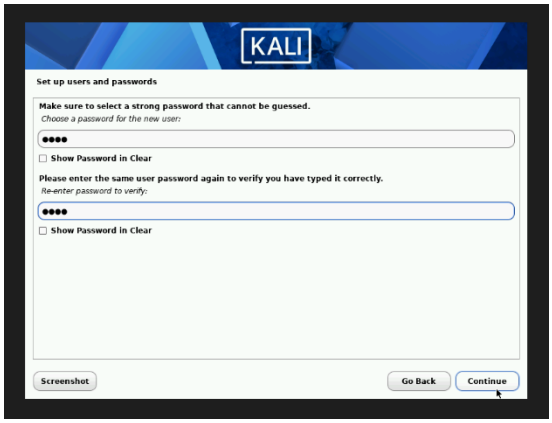
- Full Name
- Username
- Password

Choose a strong password.

Click Continue.





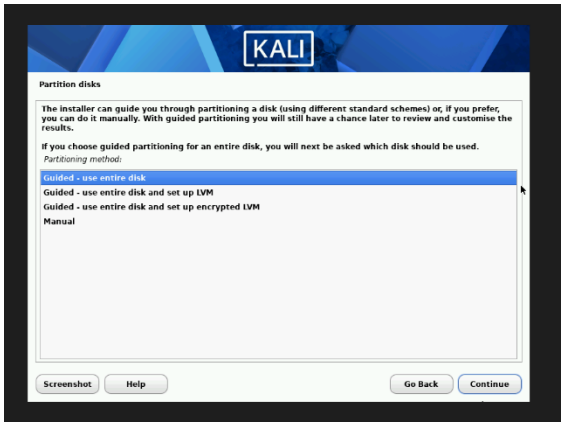


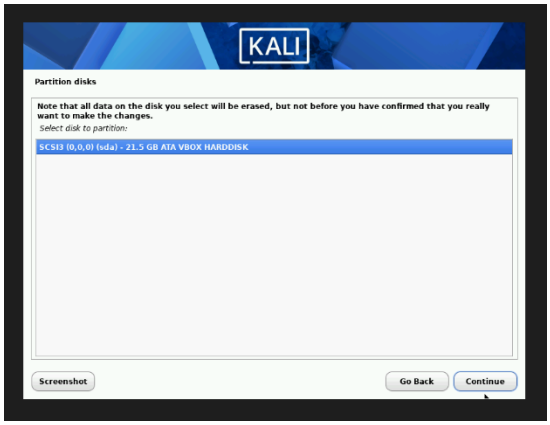
Step 7: Disk Partitioning

Select:

Guided – Use Entire Disk

Click Continue.





Select:

All Files in One Partition

Click Continue.



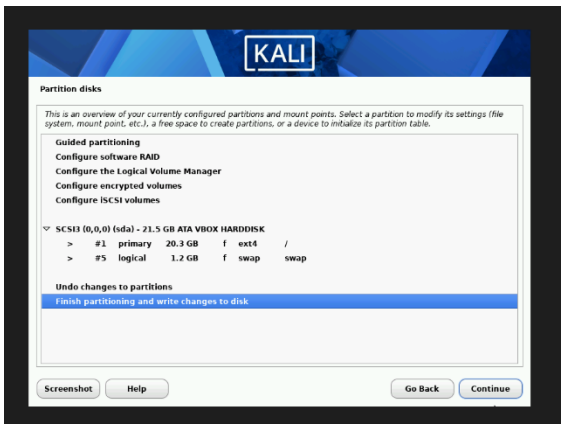
Choose:

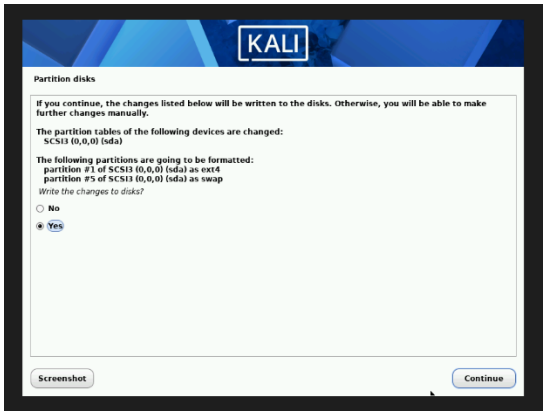
Finish Partitioning and Write Changes to Disk

Select:

Yes

Click Continue.





Step 8: Install Base System

The installer copies files and installs the operating system.

Wait until installation completes.

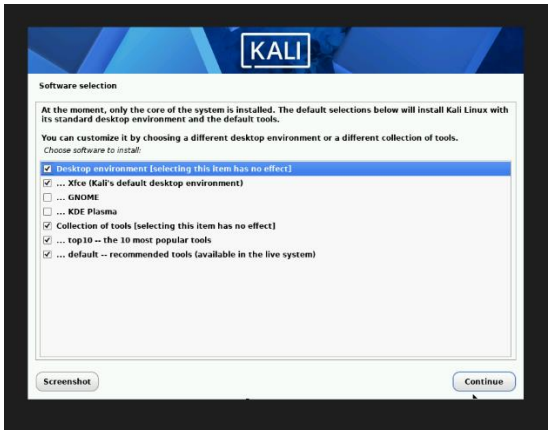
Step 9: Configure Package Manager

Select:

Yes

This allows downloading packages and updates from Kali repositories.

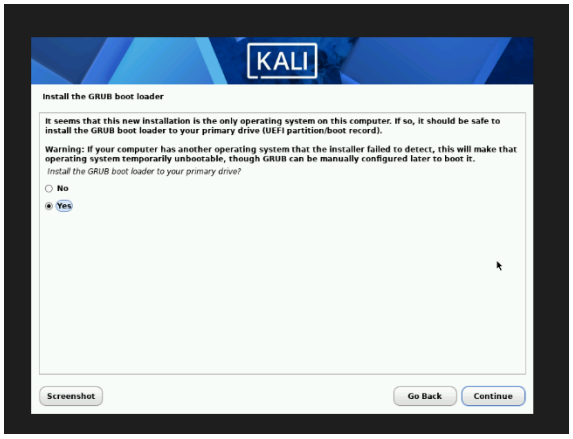
Click Continue.



Step 10: Install GRUB Boot Loader

Select:

Yes



Choose:

/dev/sda

Click Continue.



Step 11: Complete Installation

When installation finishes:

Click Continue



The virtual machine will restart automatically.

Step 12 : First Login

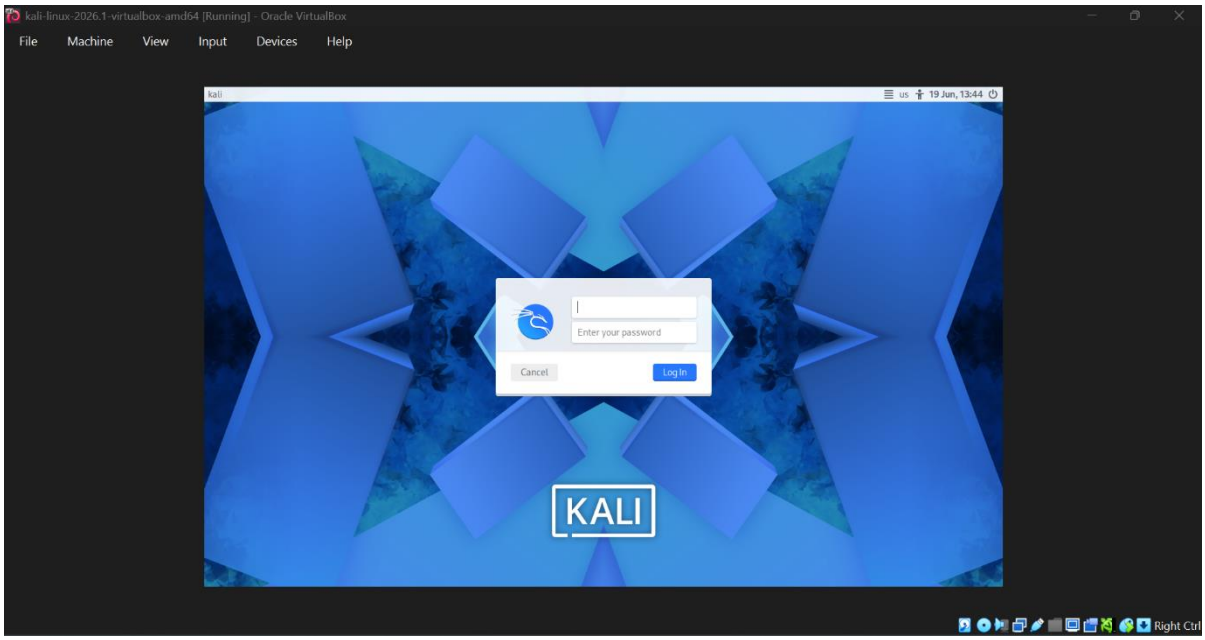
At the login screen:

Enter:

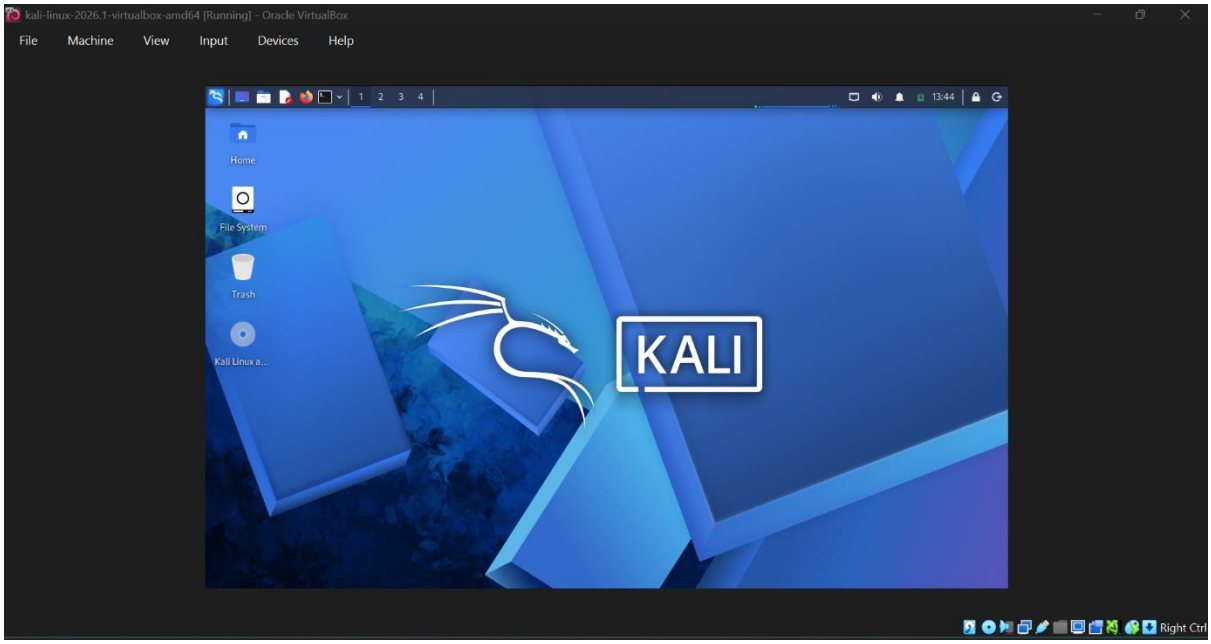
- Username
- Password

created during installation.

Click Log In.



The Kali Linux desktop environment will load.



Introduction to metasploitable :

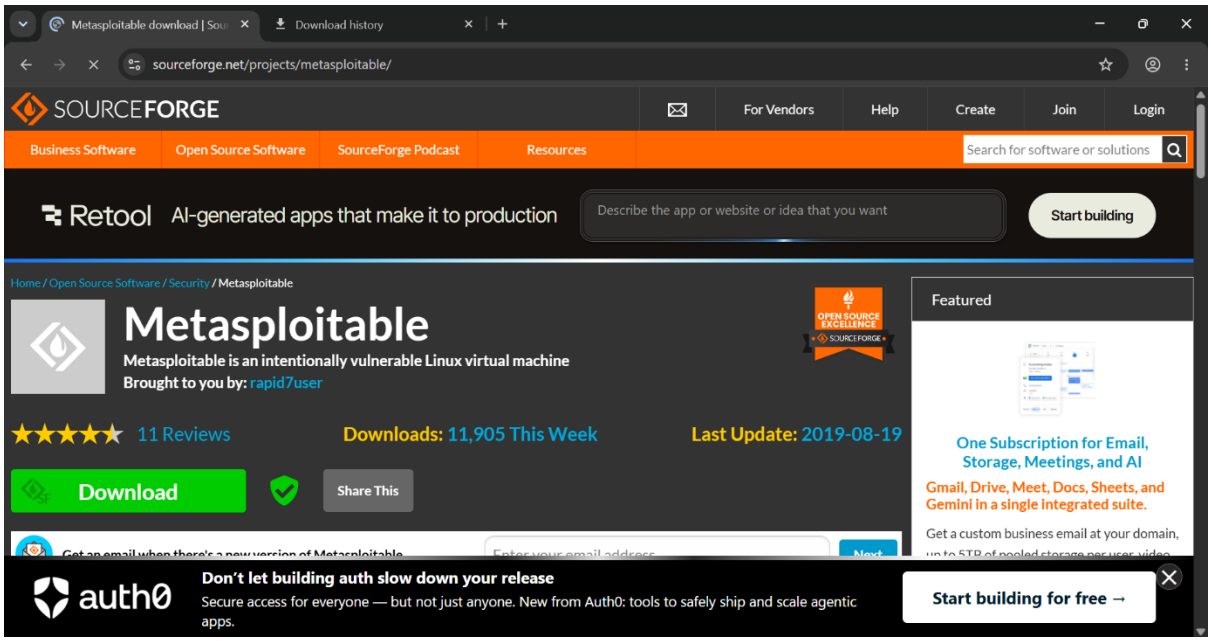
Metasploitable 2 is a deliberately vulnerable Linux virtual machine created by Rapid7 for learning and practicing penetration testing, vulnerability assessment, and ethical hacking.

It is designed to contain numerous known security vulnerabilities so that students and security professionals can safely practice identifying and exploiting them in a controlled lab environment.

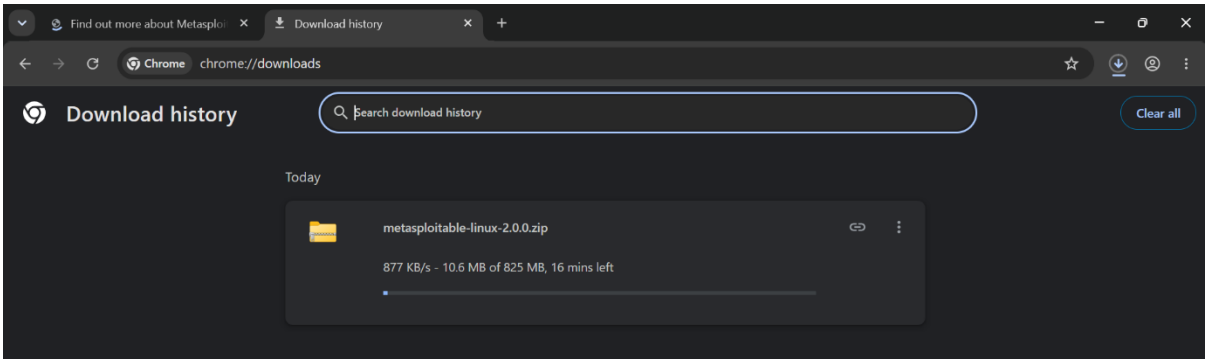
Installation and setting up metasploitable2 :

Step 1: Download Metasploitable 2

1. Visit the official project page:
    - Rapid7 Metasploitable Download



2. Download Metasploitable2-Linux.zip.



3. Extract the ZIP file.
4. Inside the extracted folder, you should see a file named something like:
    - Metasploitable.vmdk

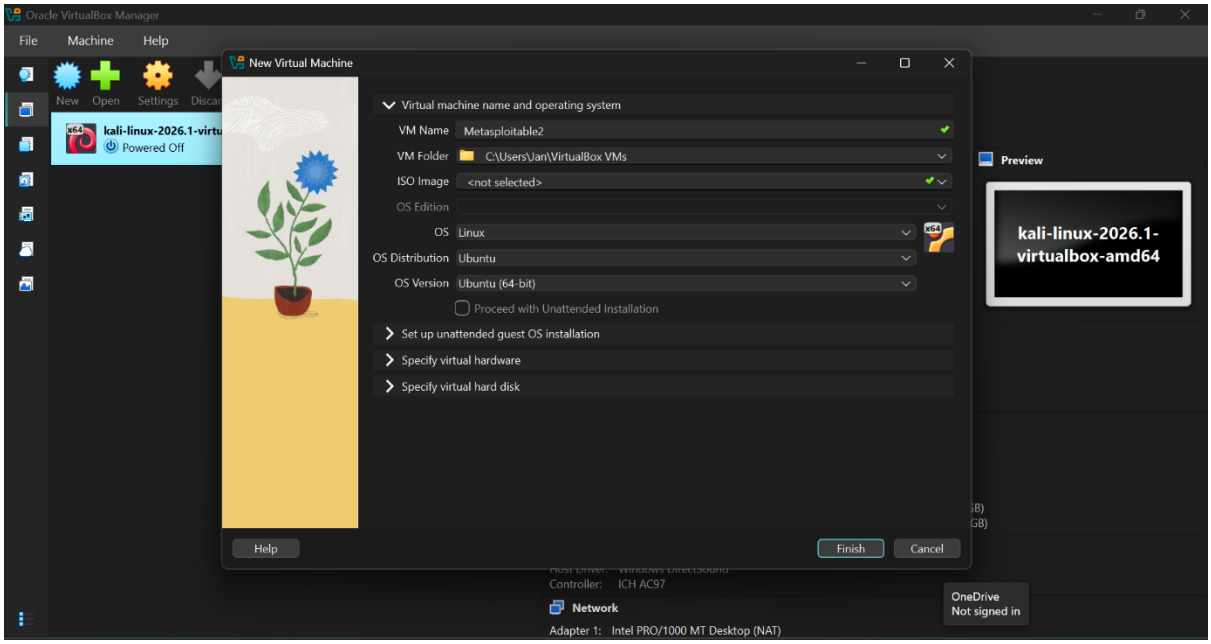
Step 2: Open VirtualBox

1. Launch VirtualBox.
2. Click New.

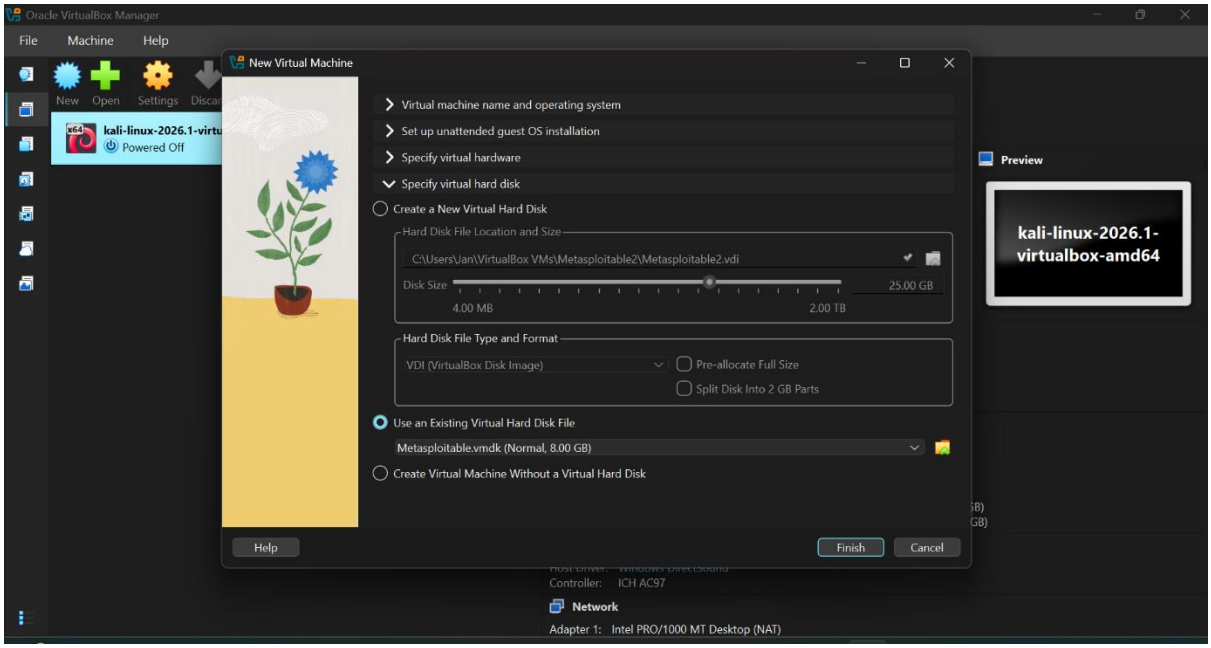
Fill in:

- Name: Metasploitable2
- Type: Linux
- Version: Ubuntu (64-bit) (or Ubuntu 32-bit if available)

Click Next.

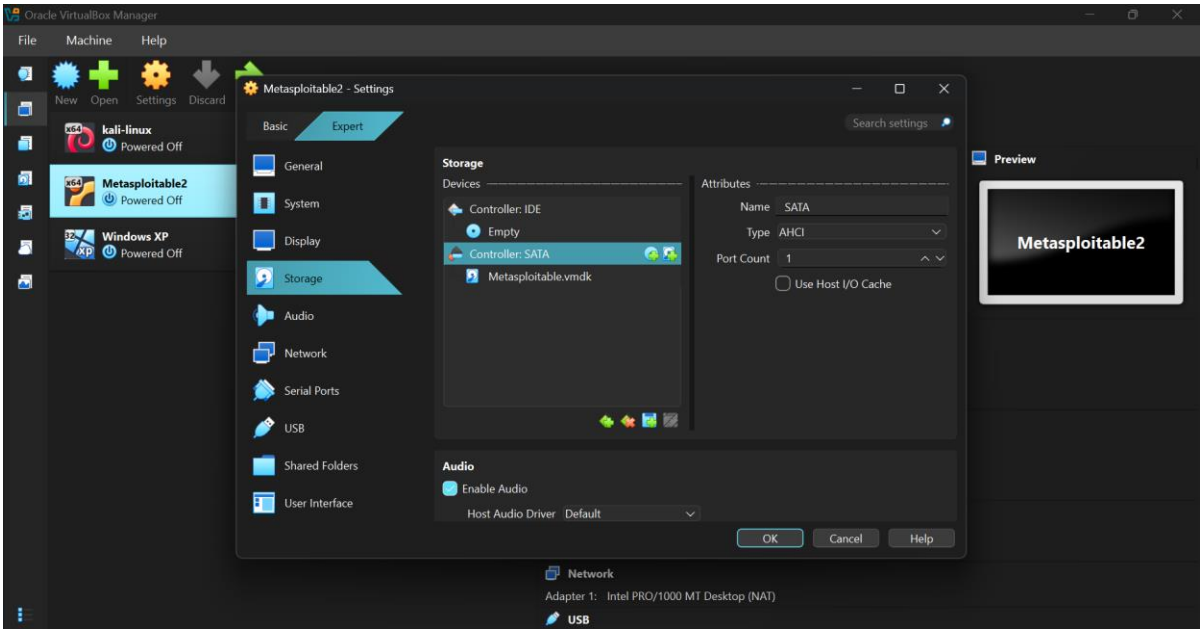


Change to Metasploitable.vmdk in existing virtual Hard Disk File.



Step 3: Use Existing Virtual Disk

1. Choose:
    - Use an Existing Virtual Hard Disk File
2. Click the folder icon.
3. Browse to the extracted Metasploitable folder.
4. Select Metasploitable.vmdk.
5. Click Create.



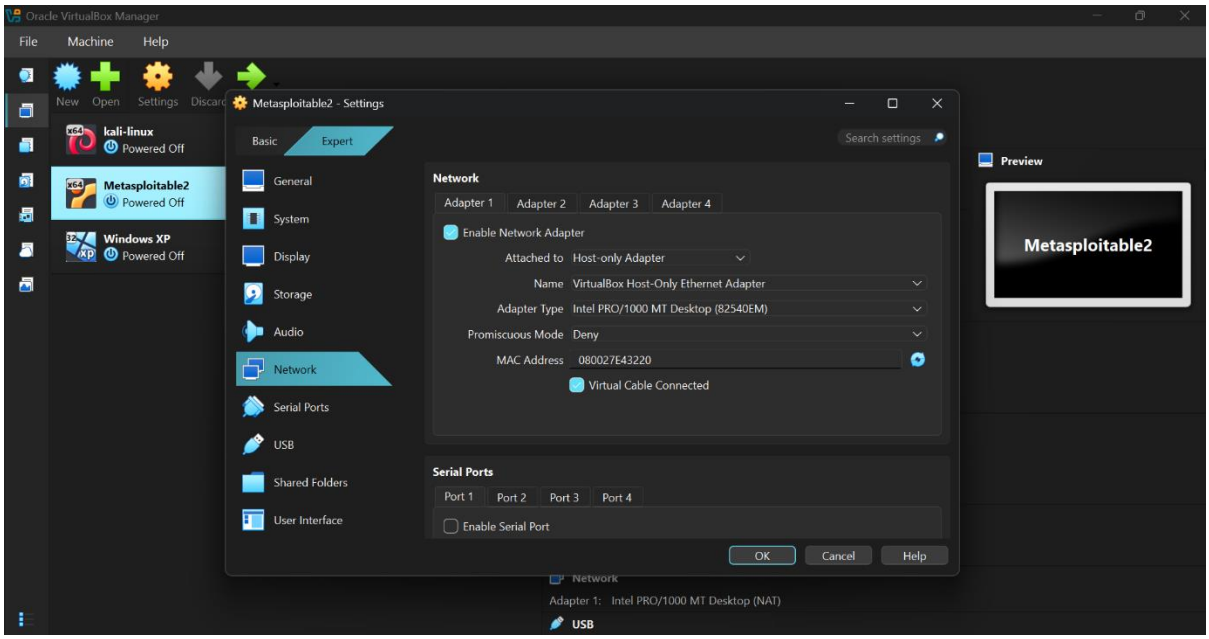
Step 4: Configure Network

1. Select the VM.
2. Click Settings → Network.

For penetration testing labs:

Adapter 1

- Enable Network Adapter
- Attached to: Host-Only Adapter

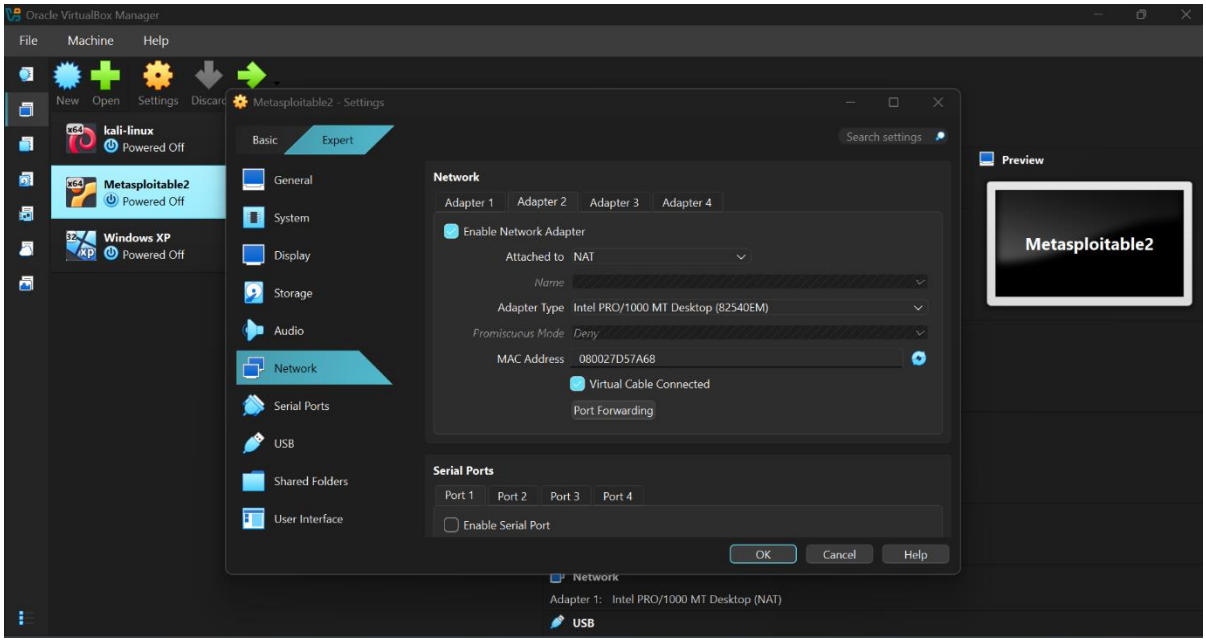


(Optional)

Adapter 2

- Enable Network Adapter
- Attached to: NAT

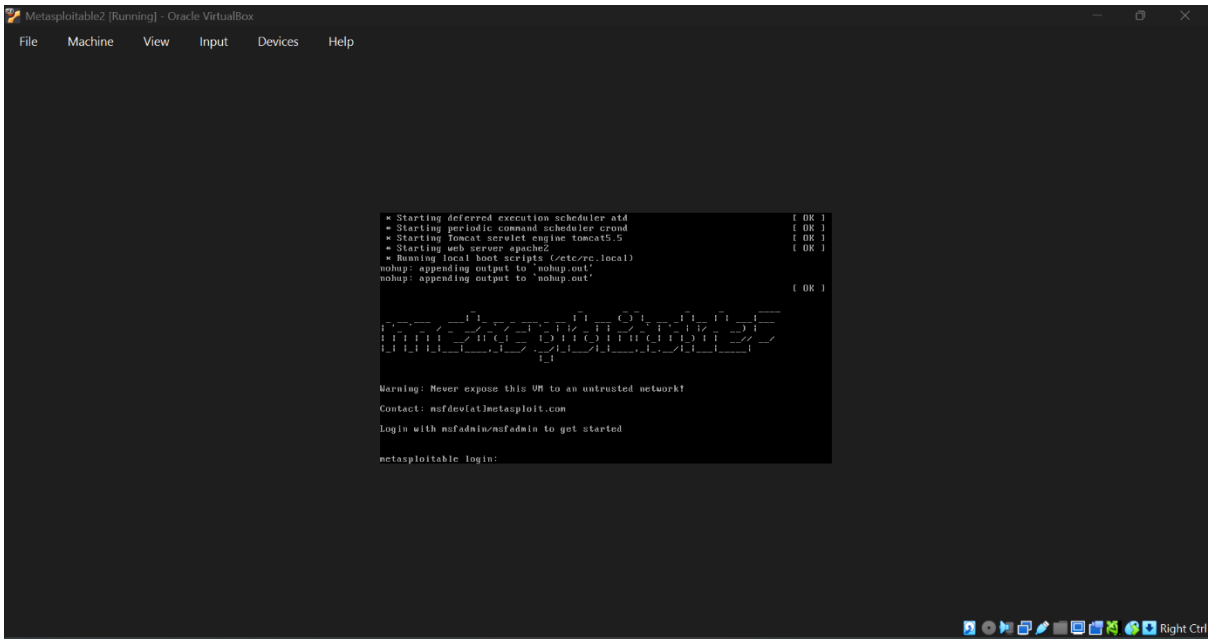
Click OK.



Step 5: Start Metasploitable

1. Click Start.
2. Wait for Linux to boot.

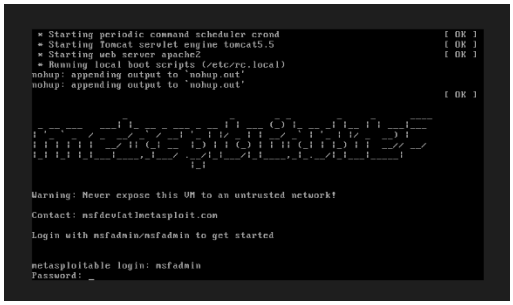
You should see a login screen.



Step 6: Login

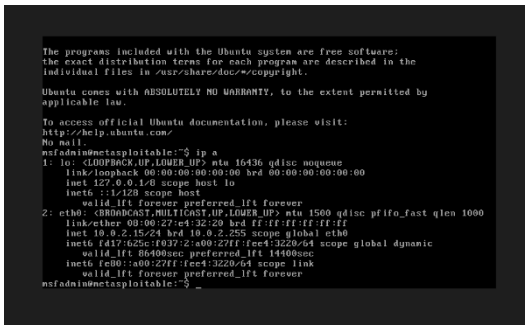
Default credentials:

Username: msfadmin
Password: msfadmin



After login:

ip a



Note the IP address (for example 192.168.56.101).

Step 7: Verify Connectivity from Kali

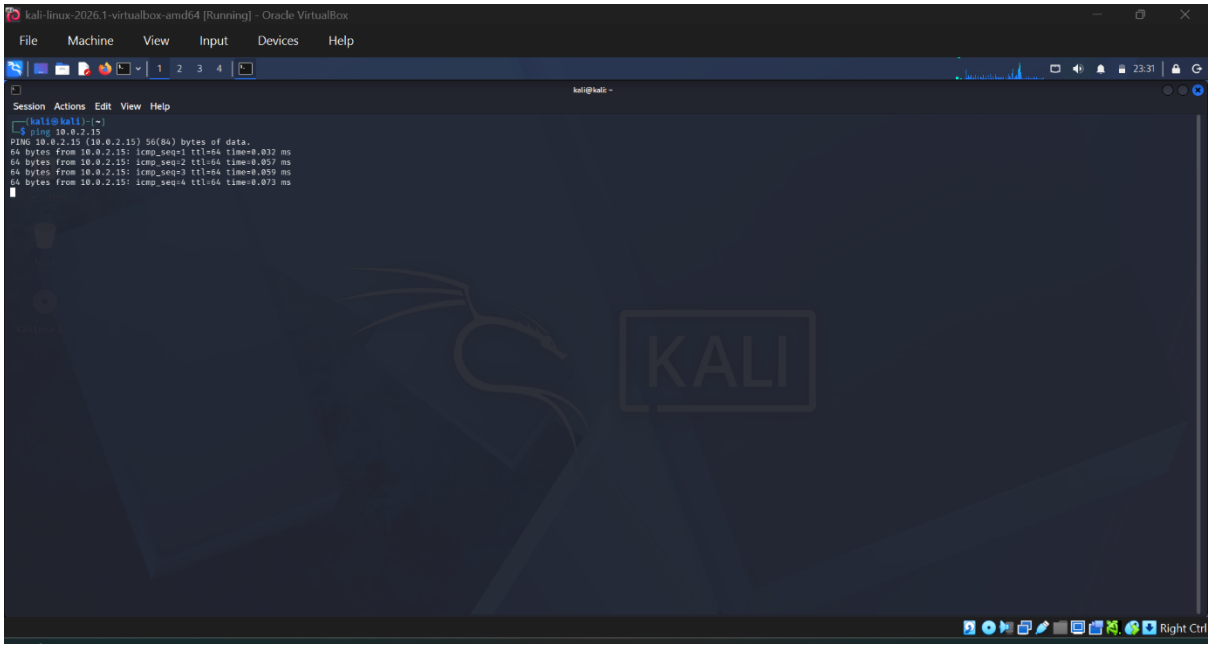
In your Kali VM:

ping <Metasploitable-IP>

Example:

ping 192.168.56.101

If replies are received, the lab is ready.



Introduction to Windows XP :

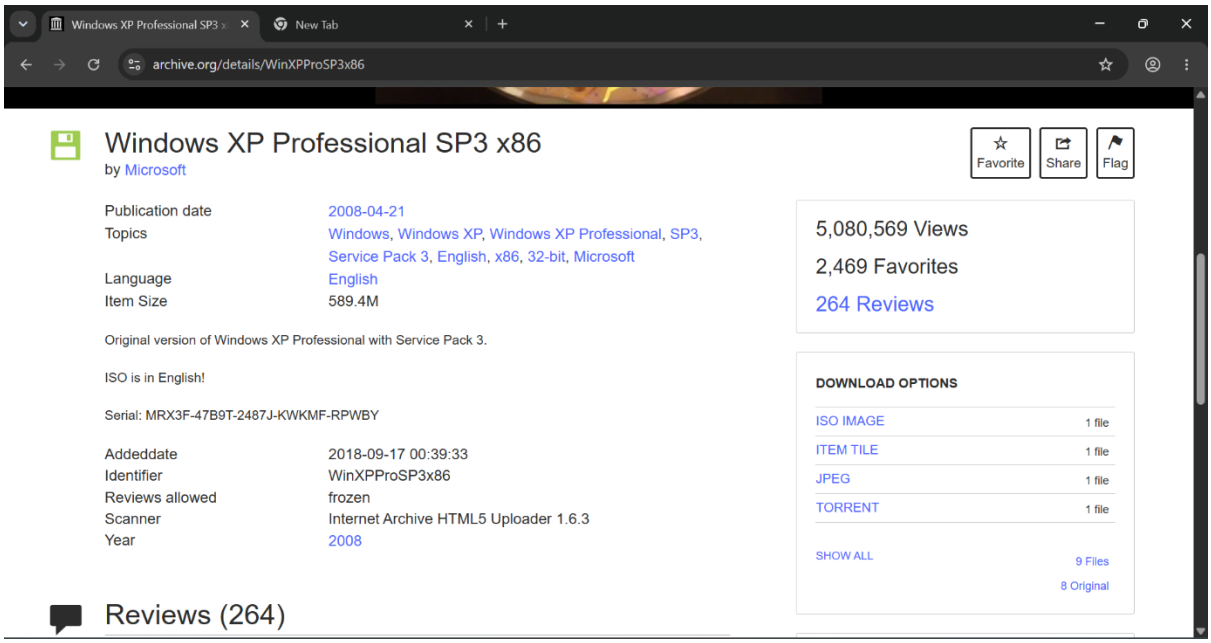
Windows XP is a desktop operating system developed by Microsoft and released on October 25, 2001. It was one of the most popular versions of Windows due to its user-friendly interface, stability, and broad hardware support.

The name XP stands for "eXPerience", reflecting Microsoft's goal of providing an improved user experience compared to earlier Windows versions.

Installation and setting up windows XP :

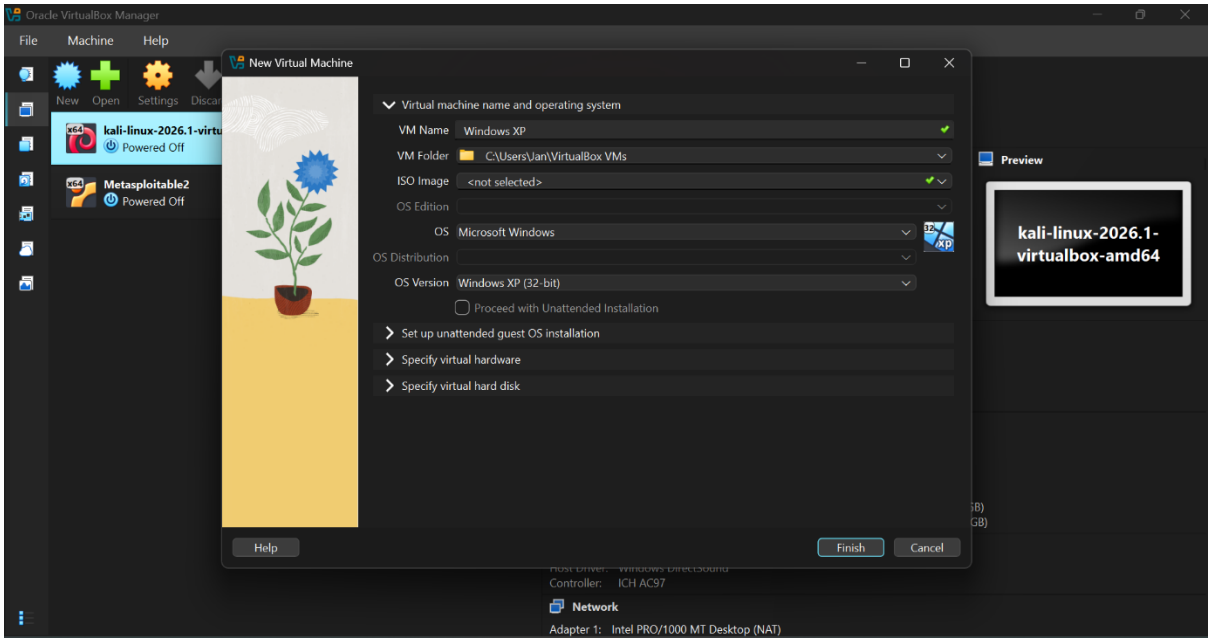
Step 1 : Download Windows XP ISO file

https://archive.org/details/WinXPProSP3x86



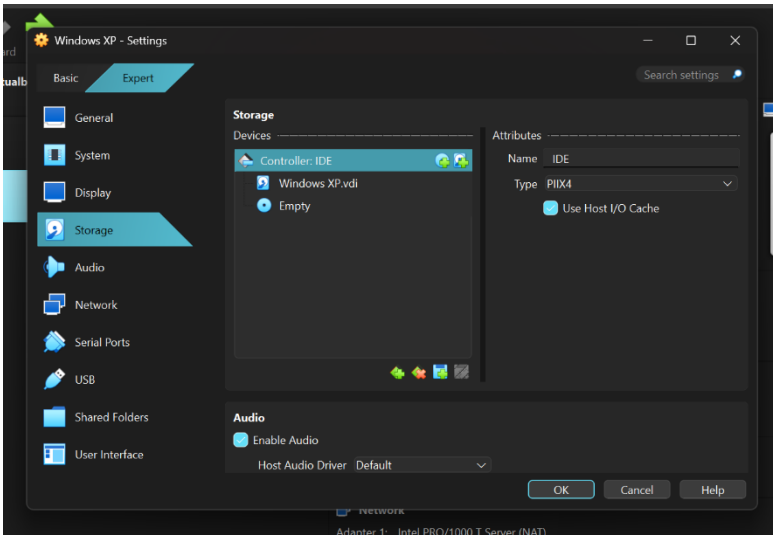
Step 2: Create a New Virtual Machine

1. Open VirtualBox.
2. Click New.
3. Name: Windows XP.
4. Type: Microsoft Windows.
5. Version: Windows XP (32-bit).
6. Click Next.



Step 3: Create Virtual Hard Disk

1. Select Create a virtual hard disk now.
2. Click Create.
3. Choose VDI (VirtualBox Disk Image).
4. Click Create.



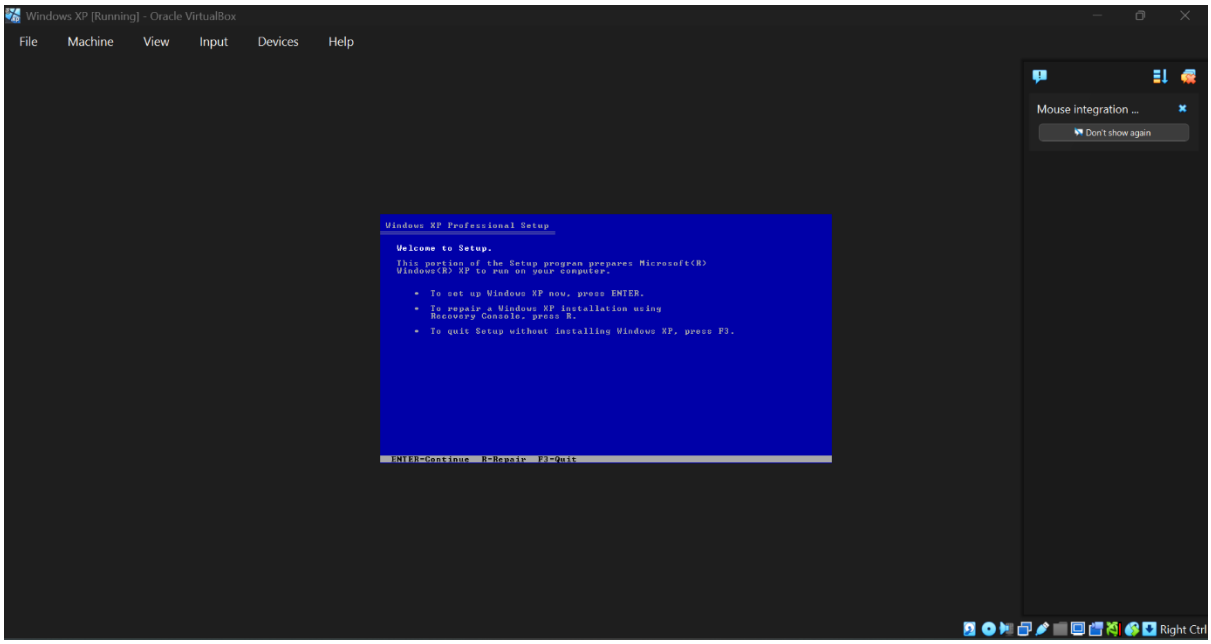
Step 4: Attach Windows XP ISO

1. Select the Windows XP VM.
2. Click Settings.
3. Go to Storage.

4. Click the empty CD icon.
5. Click the disk icon on the right.
6. Select Choose a Disk File.
7. Browse and select the Windows XP ISO.
8. Click OK.

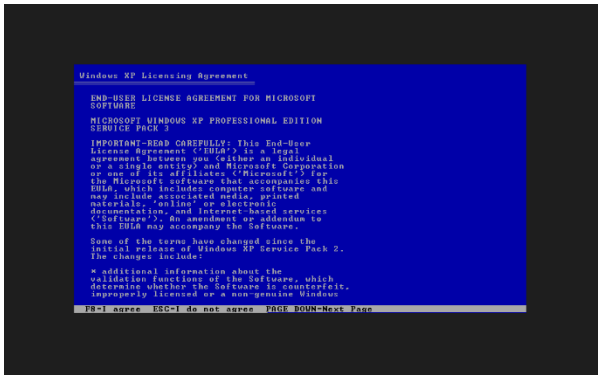
Step 5: Start Installation

1. Click Start.
2. The VM will boot from the XP ISO.
3. Setup screen appears.

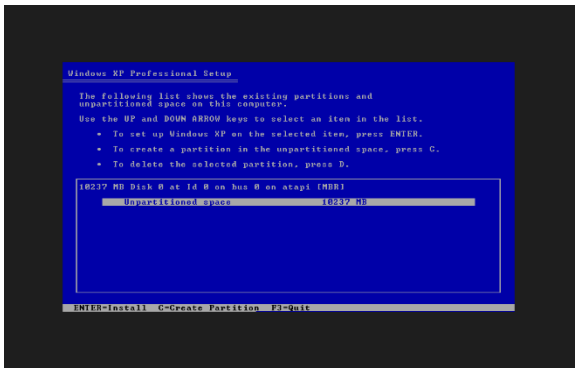


Step 6: Partition the Virtual Disk

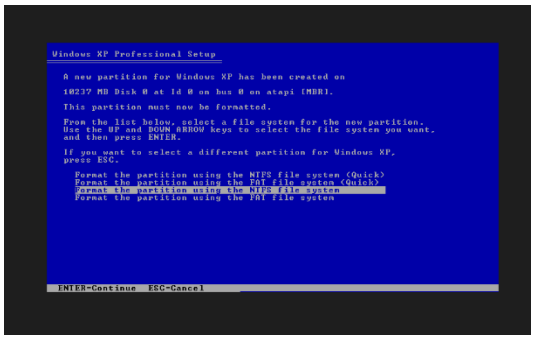
1. Press Enter to start setup.
2. Press F8 to accept the license.



3. Select Unpartitioned Space.
4. Press Enter.

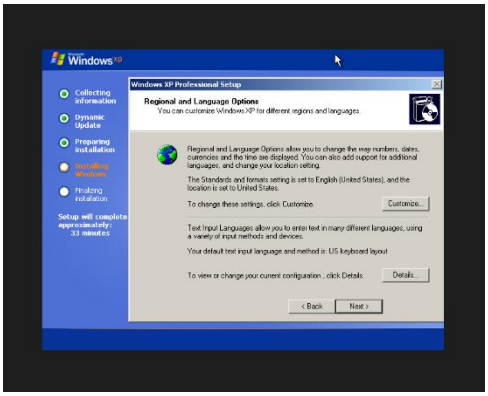


5. Choose Format using NTFS (Quick).

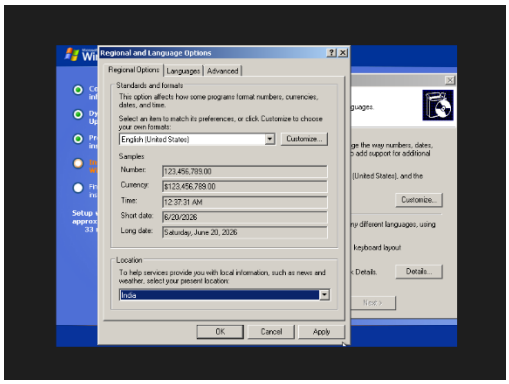


Step 7: Install Windows XP

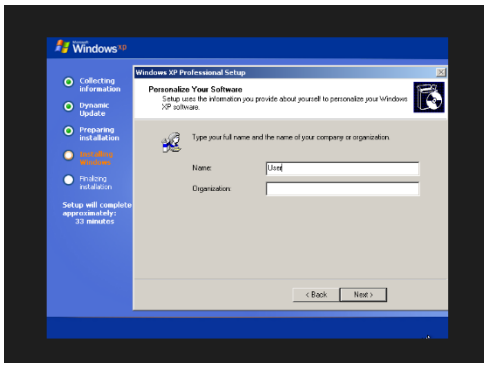
1. Files will copy.
2. VM will restart automatically.
3. Do not press any key when it reboots.
4. Follow the graphical setup:

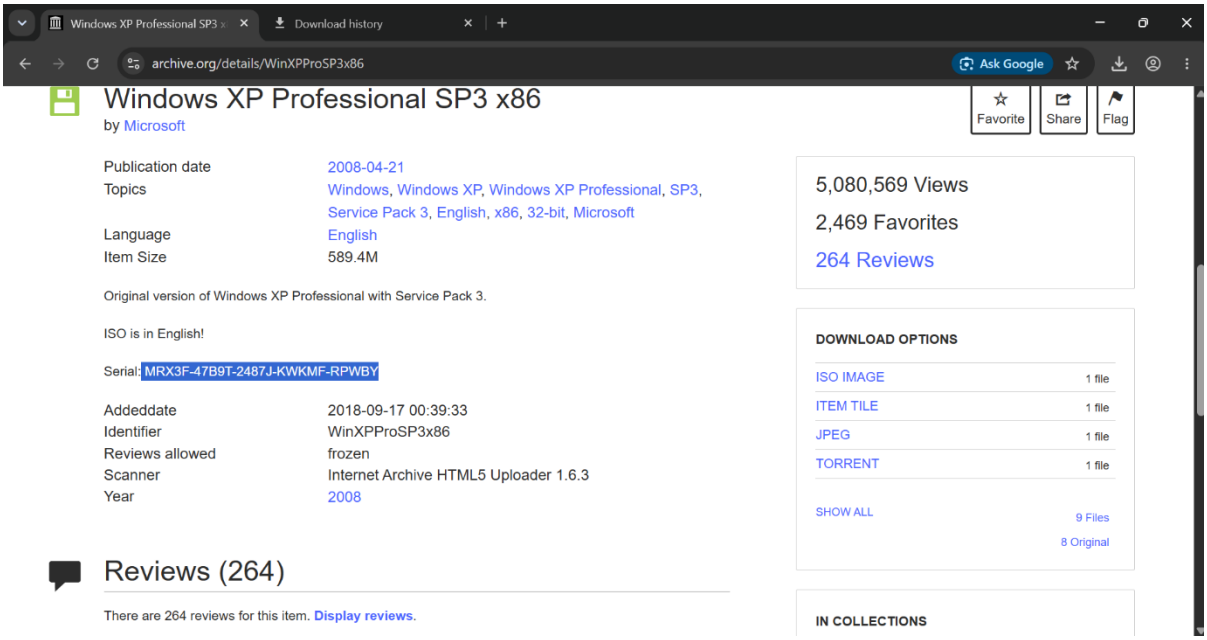


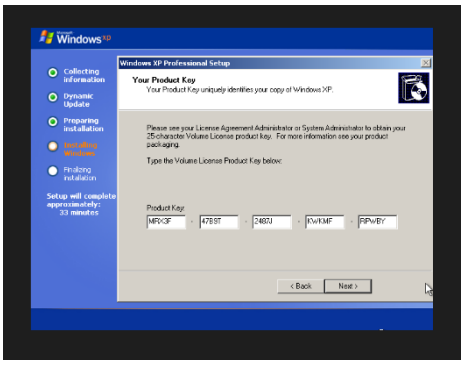
- Region and language

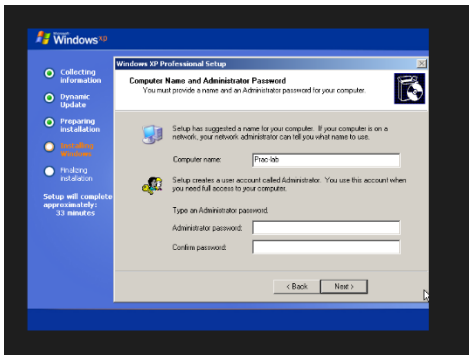


- Name
- Organization leave it blank

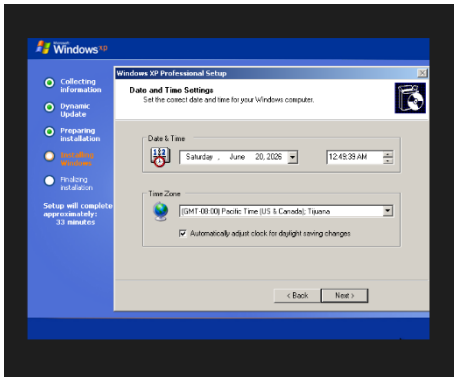


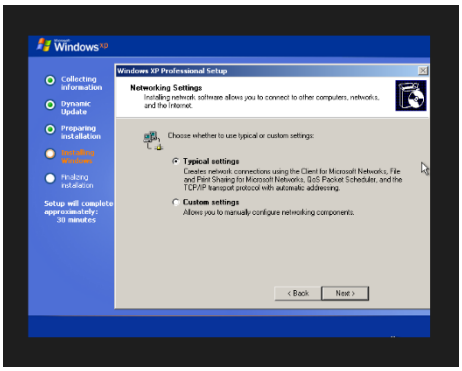


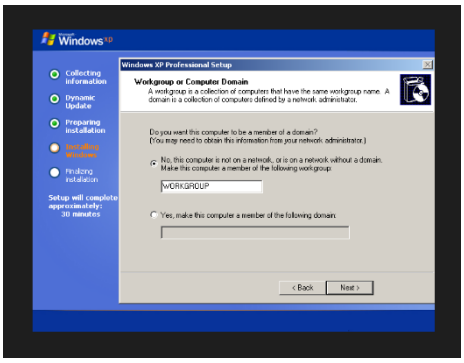




- Date and time

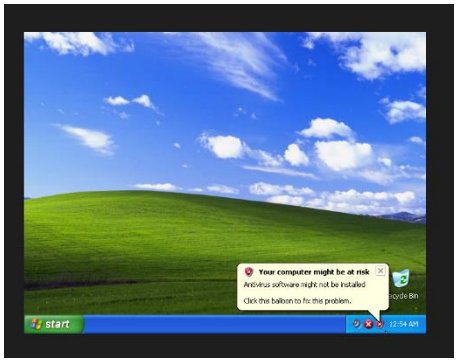






Step 8: Complete Setup

1. Windows XP will boot to the desktop.
2. Remove the ISO:
    - Devices → Optical Drives → Remove Disk.
3. Restart the VM.

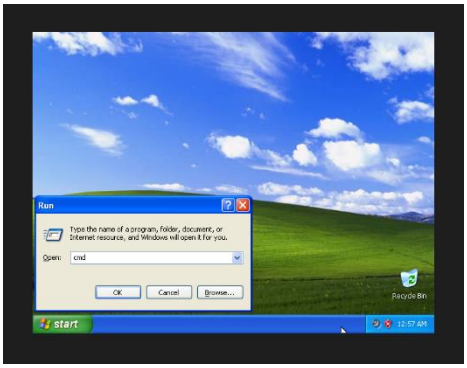


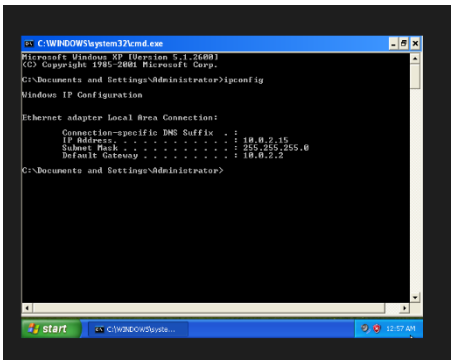






- Press Windows+R and enter cmd to open the terminal.Then enter the ipconfig to find the ip address.


Run
Type the name of a program, folder, document, or Internet resource, and Windows will open it for you.
Open:
cmd
OK
Cancel
Browse...
Recycle Bin
start
12:57 AM


C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.
C:\Documents and Settings\Administrator>ipconfig
Windows IP Configuration
Ethernet adapter Local Area Connection:
Connection-specific DNS Suffix . :
IP Address. . . . . : 10.0.2.15
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 10.0.2.2
C:\Documents and Settings\Administrator>
start
12:57 AM

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